ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ASSIGNMENT – 6

Implement Linear Regression Algorithm

Dataset: Scikit-learn Diabetes Dataset

1. Aim

To implement the Linear Regression algorithm for predicting the continuous target variable of the Diabetes dataset and evaluate the regression model using standard error metrics.

2. Objectives

- Understand the concept of supervised regression.
- Fit a multiple linear regression model using ten independent variables.
- Split the data into training and testing sets.
- Evaluate predictions using MAE, MSE, RMSE and R^2.
- Analyze feature coefficients and residuals.

3. Dataset Description

The Scikit-learn Diabetes dataset contains 442 observations, 10 independent features and one continuous target representing a quantitative measure of disease progression. The target is used directly without converting it into classes.

Property	Value
Samples	442
Features	10
Target type	Continuous
Training samples	353
Testing samples	89

4. Theory

Linear Regression is a supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. Multiple linear regression extends the simple linear model to several predictors.

The general form is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_p x_p + \varepsilon$, where β_0 is the intercept, β_i are coefficients, x_i are features and ε represents the error term.

The coefficients describe the expected change in the predicted target associated with a one-unit change in a feature while holding the other included features constant. The sign indicates the direction of the association.

Ordinary Least Squares estimates coefficients by minimizing the sum of squared residuals. A residual is the difference between the observed target and the model's predicted target.

MAE is the average absolute prediction error and is easy to interpret because it uses the same units as the target. MSE is the average squared error and penalizes large errors more strongly.

RMSE is the square root of MSE and is also expressed in the target's units. R^2 measures the proportion of target variance explained by the model relative to a baseline model using the target mean.

Actual-versus-predicted plots provide a visual check of prediction quality. Points close to the diagonal reference line indicate predictions close to the observed values.

Residual plots are useful for diagnosing systematic errors. Ideally, residuals should be distributed around zero without obvious patterns, although real datasets may not satisfy every ideal regression assumption.

Regression is different from correlation: a regression model estimates a predictive relationship, while correlation measures the strength of linear association between two variables.

Linear regression relies on assumptions such as linearity, independent errors, approximately constant error variance and, for some inferential procedures, normally distributed errors. Prediction can still be useful when these assumptions are not perfectly satisfied, but diagnostic plots should be considered.

Multiple regression also introduces the possibility of multicollinearity, where predictors are strongly related to each other. Multicollinearity can make individual coefficients unstable even when overall predictive performance is reasonable.

The train-test split provides an estimate of generalization performance. In this experiment, 353 observations are used for training and 89 are held out for testing.

5. Methodology

- Load the Diabetes dataset.
- Separate the ten features and continuous target.
- Split the dataset into 80% training and 20% testing sets.
- Fit a LinearRegression model on the training data.
- Generate predictions for the test data.
- Calculate MAE, MSE, RMSE and R^2.
- Inspect the feature coefficients.
- Plot actual versus predicted values, residuals, coefficient magnitudes and sample test predictions.

6. Outputs

6.1 Model Metrics

Metric	Value
Mean Absolute Error (MAE)	42.7941
Mean Squared Error (MSE)	2900.1936
Root Mean Squared Error (RMSE)	53.8534
R2 Score	0.4526

6.2 Feature Coefficients

Feature	Coefficient
age	37.9040
sex	-241.9644
bmi	542.4288
bp	347.7038
s1	-931.4888
s2	518.0623
s3	163.4200
s4	275.3179
s5	736.1989
s6	48.6707

6.3 Actual vs Predicted

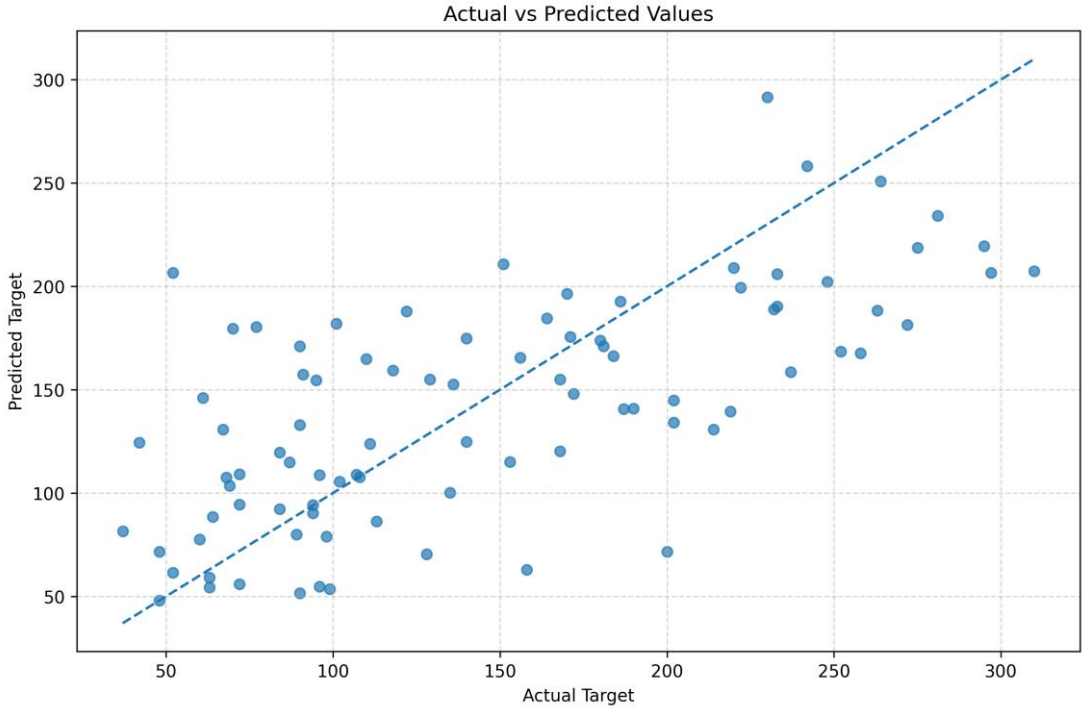


Figure 1: Actual versus predicted target values

6.4 Residual Plot

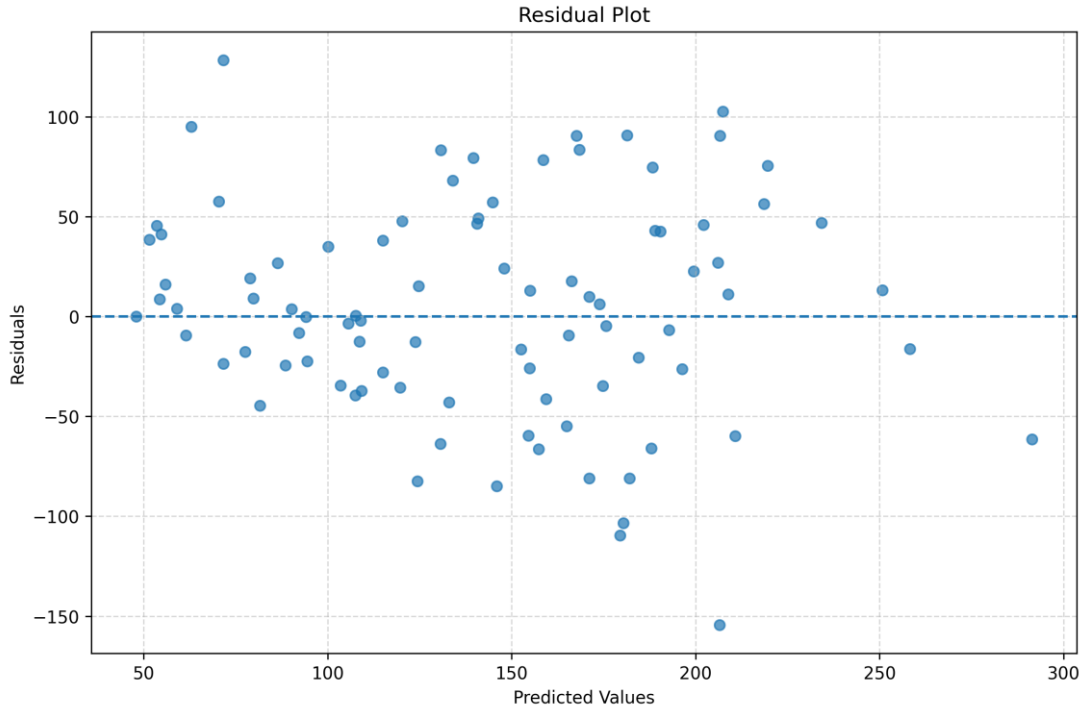


Figure 2: Regression residual plot

6.5 Feature Coefficients

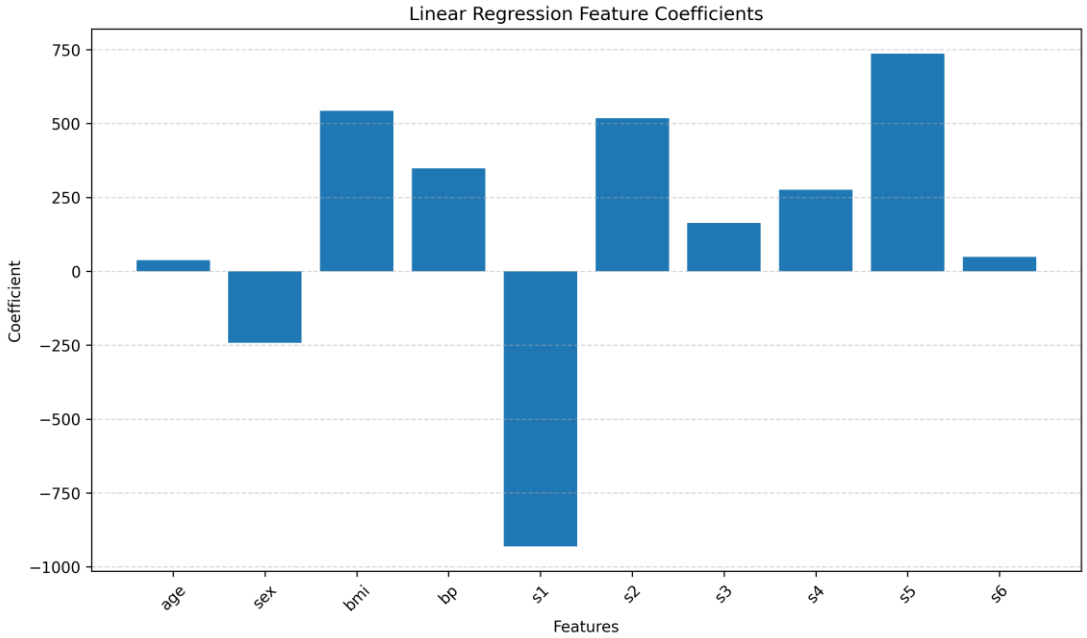


Figure 3: Linear regression feature coefficients

6.6 Test Actual vs Predicted Samples

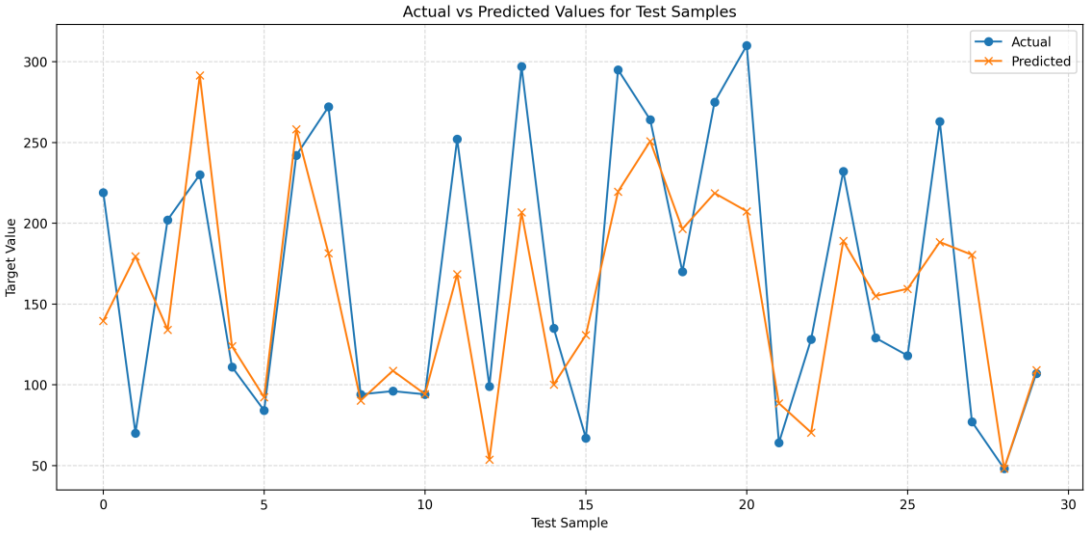


Figure 4: Actual and predicted values for test samples

7. Result

Linear Regression achieved MAE = 42.7941, MSE = 2900.1936, RMSE = 53.8534 and R^2 = 0.4526 on the supplied test results. The model therefore explains approximately 45.26% of the variation in the test target relative to the baseline.

8. Observations

- The model uses all ten independent features.
- The R^2 score is 0.4526.
- The RMSE is 53.8534 target units.
- The coefficient plot shows differences in the estimated contribution of features.
- The actual-versus-predicted and residual plots provide visual information about model errors.

9. Conclusion

Linear Regression was successfully implemented for continuous target prediction. The experiment demonstrates model fitting, quantitative error evaluation and graphical residual analysis. The results provide a baseline against which nonlinear or regularized regression models can be compared.